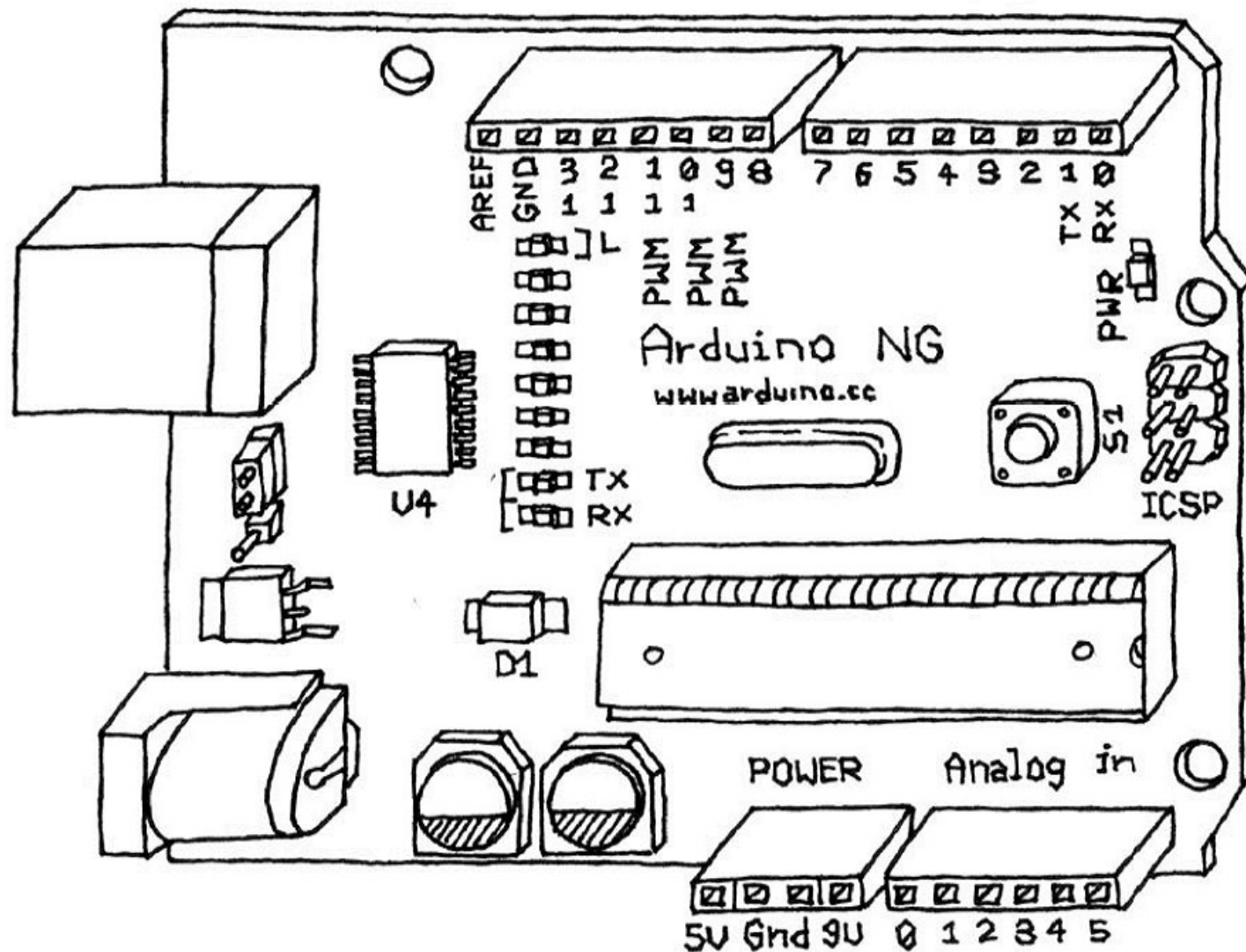




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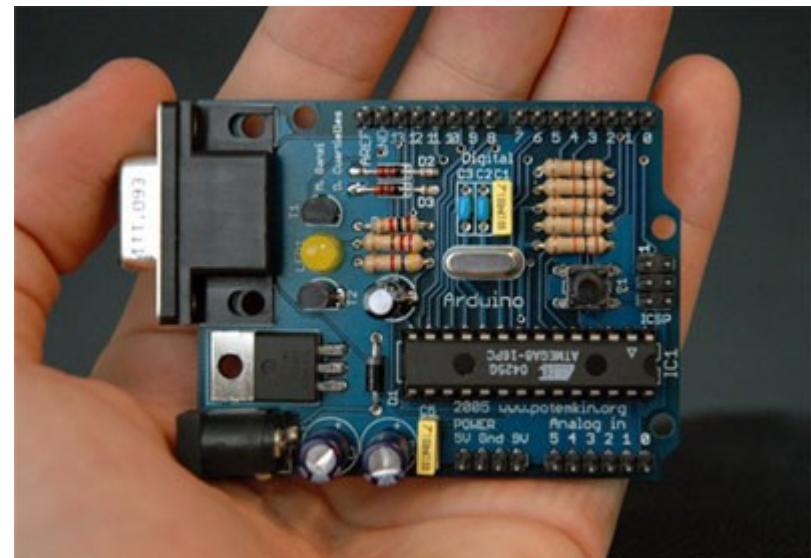
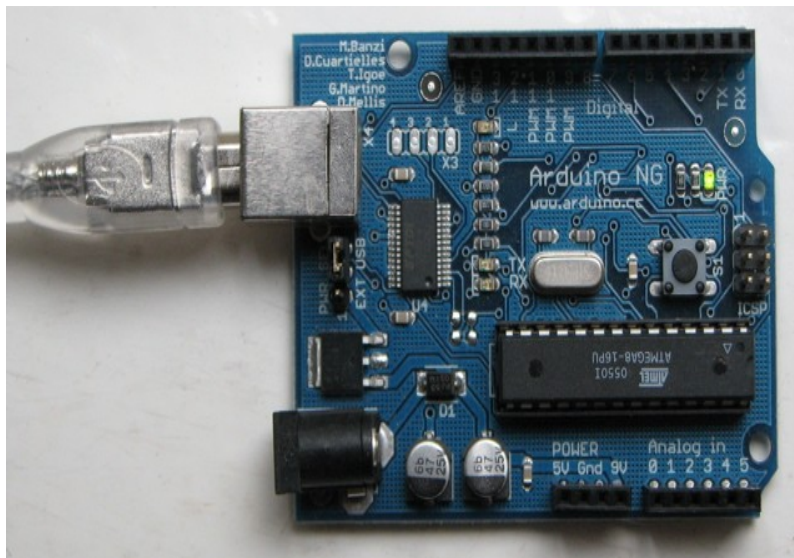
//overview

- Arduino Workshop
 - Introduction
 - Arduino Board
 - Arduino Language
 - Arduino Programming Environment
 - Example: PushButton
 - Tips & Tricks for embedded Programming
 - Resources
- Uebung
 - Aufgabe 3
 - Aufgabe 2 – Showcase
 - Ausblick

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//introduction Arduino

- open-source electronics prototyping platform based on flexible, easy-to-use hardware and software (platform for physical computing)
- intended for artists, designers, hobbyists, and anyone interested in creating interactive objects or environments
- can be used to implement self containing/embedded applications/setups or applications that communicate with a pc or other computer
- components:
 - Arduino Board (comes in different variations)
 - Arduino Programming Language (based on Wiring / AVR Libc)
 - Arduino Programming Environment (based on Processing)



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//Arduino Board/Microcontroller

Microcontroller: single integrated circuit, commonly with the following features:

- central processing unit - ranging from small and simple 4-bit processors to complex 32- or 64-bit processors
- discrete input and output bits, allowing control or detection of the logic state of an individual package pin
- serial input/output such as serial ports (UARTs)
- other serial communications interfaces like I²C, Serial Peripheral Interface, ...
- peripherals such as timers, event counters, PWM generators, and watchdog
- volatile memory (RAM) for data storage
- ROM, EPROM, EEPROM or Flash memory for program and operating parameter storage
- clock generator
- many include analog-to-digital converters
- in-circuit programming and debugging support

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//Arduino Board/Microcontroller

Arduino Diecimila based on AVR's Atmega168:

- 14 Digital I/O Pins (of which 6 provide PWM output)
- 6 Analog/Digital Input Pins
- 16 KB Flash Memory(of which 2 KB used by bootloader)
- 1KB SRAM
- 512 bytes EEPROM
- Clock Speed: 16 MHz



+ other components on board:

- FTDI FT232 USB to serial UART interface & USB connector
- ICSP connector
- some power supply circuitry

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//Arduino Board/Microcontroller

Pin Mapping Atmega8/168 to Arduino



Arduino Pin Mapping

www.arduino.cc

digital pin 0 (RX)	(RESET) PC6	1	28	PC5 (ADC5/SCL)	analog input 5
digital pin 1 (TX)	(RXD) PD0	2	27	PC4 (ADC4/SDA)	analog input 4
digital pin 2	(TXD) PD1	3	26	PC3 (ADC3)	analog input 3
digital pin 3	(INT0) PD2	4	25	PC2 (ADC2)	analog input 2
digital pin 4	(INT1) PD3	5	24	PC1 (ADC1)	analog input 1
	(XCK/T0) PD4	6	23	PC0 (ADC0)	analog input 0
	VCC	7	22	GND	
	GND	8	21	AREF	
	(XTAL1/TOSC1) PB6	9	20	AVCC	
digital pin 5	(XTAL2/TOSC2) PB7	10	19	PB5 (SCK)	digital pin 13 (LED)
digital pin 6	(T1) PD5	11	18	PB4 (MISO)	digital pin 12
digital pin 7	(AIN0) PD6	12	17	PB3 (MOSI/OC2)	digital pin 11 (PWM)
digital pin 8	(AIN1) PD7	13	16	PB2 (SS/OC1B)	digital pin 10 (PWM)
	(ICP1) PB0	14	15	PB1 (OC1A)	digital pin 9 (PWM)

ATmega8

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//Arduino Programming Language

- Arduino sketches are C/C++ based >> operators, program control, data types / structs / class definitions, etc. as in C/C++
- The Arduino language comes from Wiring, which builds upon Processing >> structure is very similar to Processing (setup(), loop(), ...)
- compiled with the open-source compiler avr-gcc (freeware C compiler (and assembler) made available through the GNU project)
- linked against the open-source AVR Libc (high quality C library for use with GCC on Atmel AVR microcontrollers) >> you can use AVR Libc compatible code/libraries
- core: functions for
 - digital/analog input/output
 - serial communication
 - time
 - interrupts
- libraries for
 - controlling stepper motors
 - reading and writing to EEPROM ("permanent" storage)
 - software implementation of serial protocol
 - controlling led matrices
 - ...

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//Arduino Programming Language/example

```
// LED connected to digital pin 13
int ledPin = 13;

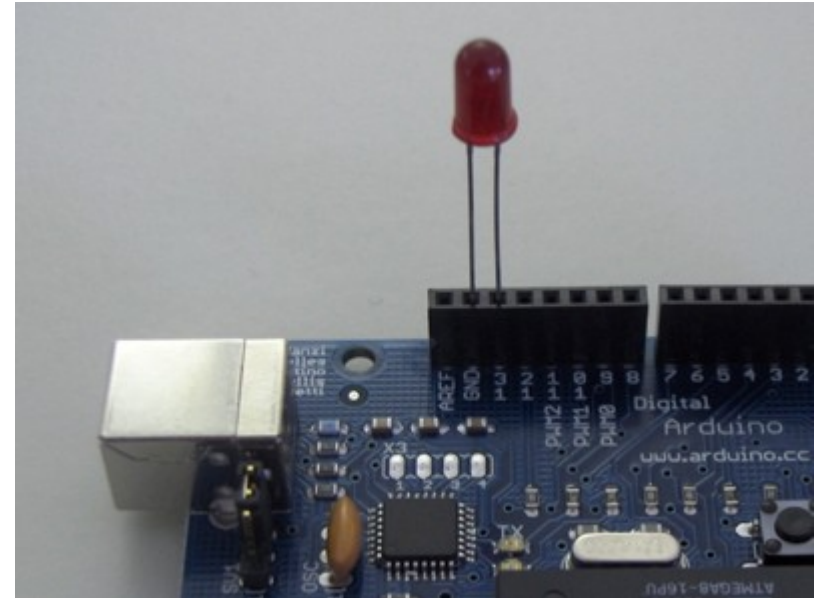
void setup()
{
  // sets the digital pin as output
  pinMode(ledPin, OUTPUT);
}

void loop()
{
  // switch the LED on
  digitalWrite(ledPin, HIGH);

  // waits for a second
  delay(1000);

  // switch the led off
  digitalWrite(ledPin, LOW);

  // waits for a second
  delay(1000);
}
```



note: the Arduino has a 220 Ohm resistor built in at pin 13, so you can easily connect a led there for debugging, etc. without burning it

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//Arduino Programming Language

structure of an Arduino
program:

```
int main(void)
{
    init();

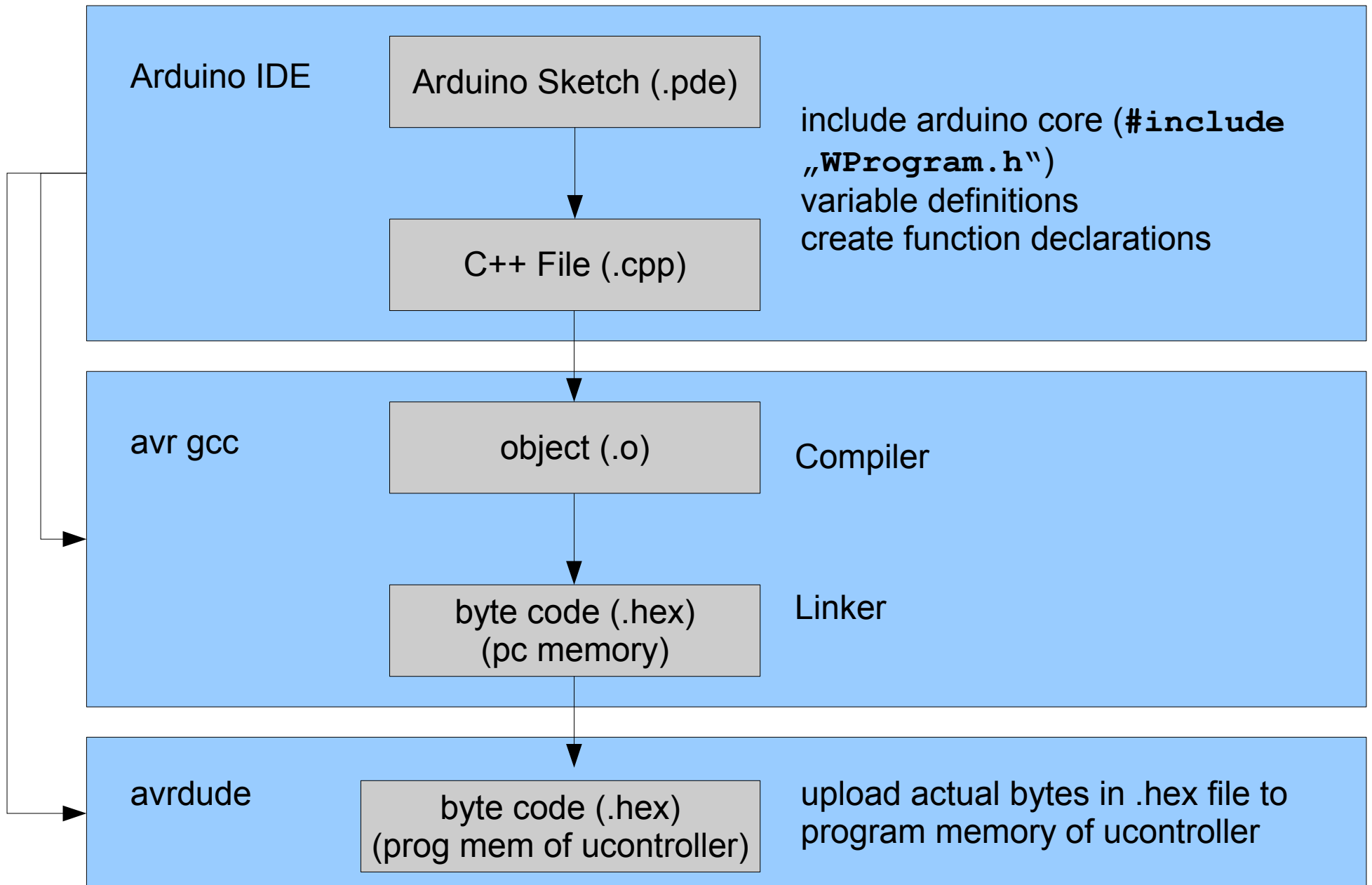
    setup();

    for (;;)
        loop();

    return 0;
}
```

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//Arduino/Programming Language/Build Cycle & Upload



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//Arduino Programming Language

```
#include "WProgram.h"
```

```
void setup();  
void loop();
```

```
// LED connected to digital pin 13  
int ledPin = 13;
```

```
void setup()  
{  
  // sets the digital pin as output  
  pinMode(ledPin, OUTPUT);  
}
```

```
void loop()  
{  
  // switch the LED on  
  digitalWrite(ledPin, HIGH);  
  
  // waits for a second  
  delay(1000);  
  
  // switch the led off  
  digitalWrite(ledPin, LOW);  
  
  // waits for a second  
  delay(1000);  
}
```

```
:1000000000C945A000C9475000C9475000C947500B7  
:100010000C9475000C9475000C9475000C9475008C  
:100020000C9475000C9475000C9475000C9475007C  
:100030000C9475000C9475000C9475000C9475006C  
:100040000C949F000C9475000C9475000C94750032  
:100050000C9475000C9475000C9475000C9475004C  
:100060000C9475000C94750000000007000201005C  
:1000700000030406000000000000000000102040864  
:100080001020408001020408102001020408102002  
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:1001100060E0809100010E94BC0168EE73E080E025  
:1001200090E00E94DC000895CFEFD4E0DEBFCDBFA9  
:100130000E940B010E9477000E947D00FDCF1F925C  
:100140000F920FB60F9211248F939F93AF93BF938B  
:100150008091020190910301A0910401B0910501E9  
:100160000196A11DB11D8093020190930301A093FC  
:100170000401B0930501BF91AF919F918F910F90B2  
...
```

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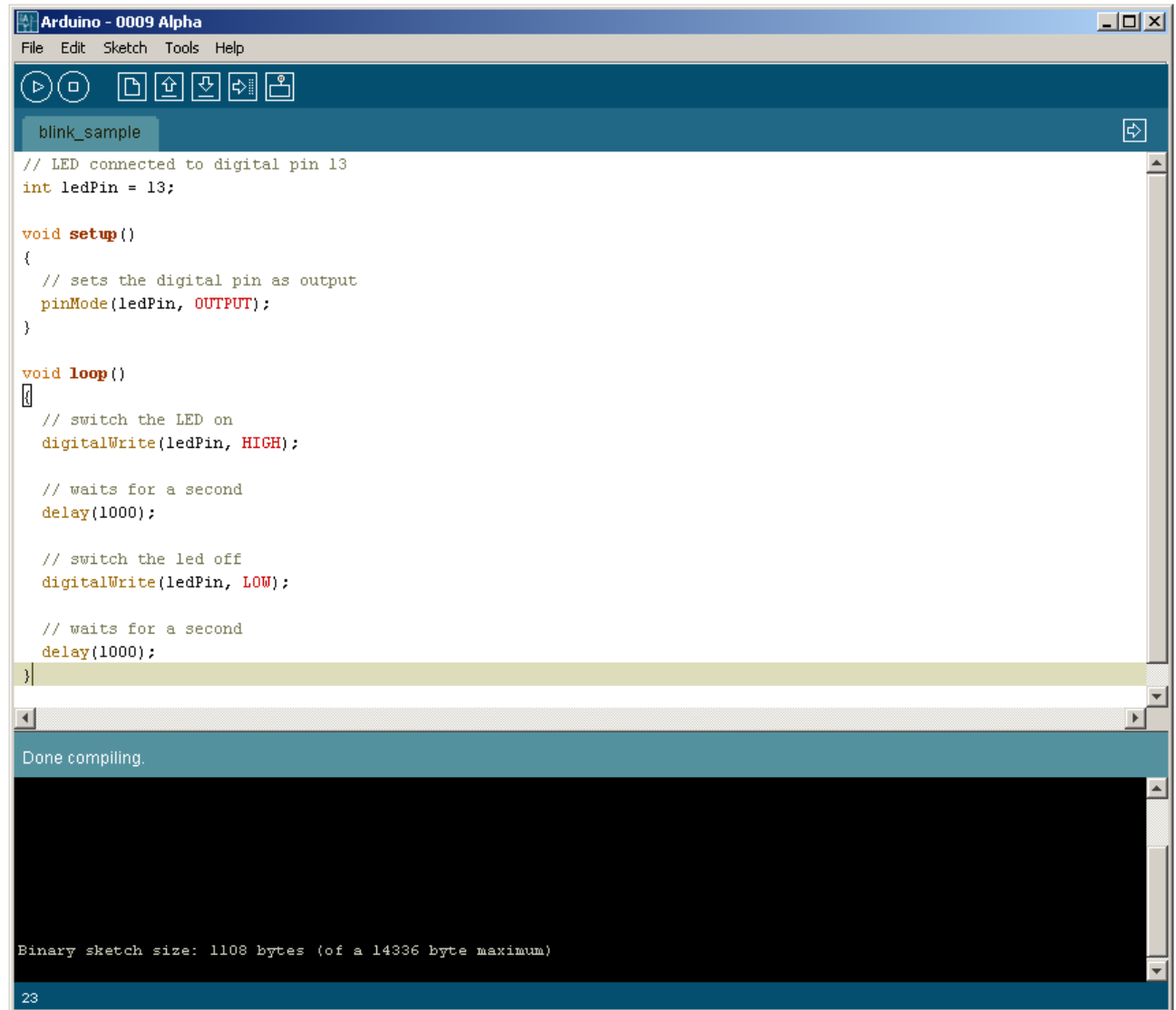
//Arduino Programming Environment

built upon Processing, looks pretty much the same, just with different colors, some buttons mean different things (verify, upload)

buttons: compile, stop, create new file, open, save, upload program & serial monitor

select microcontroller & serial port

...



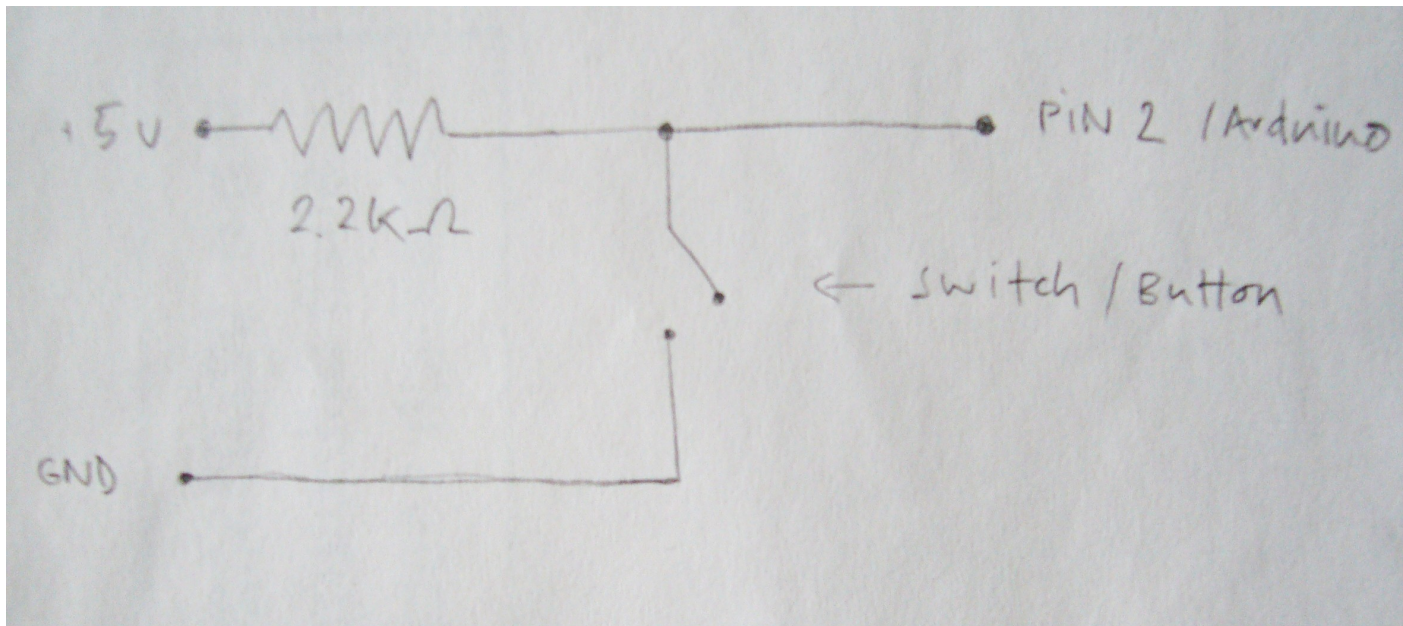
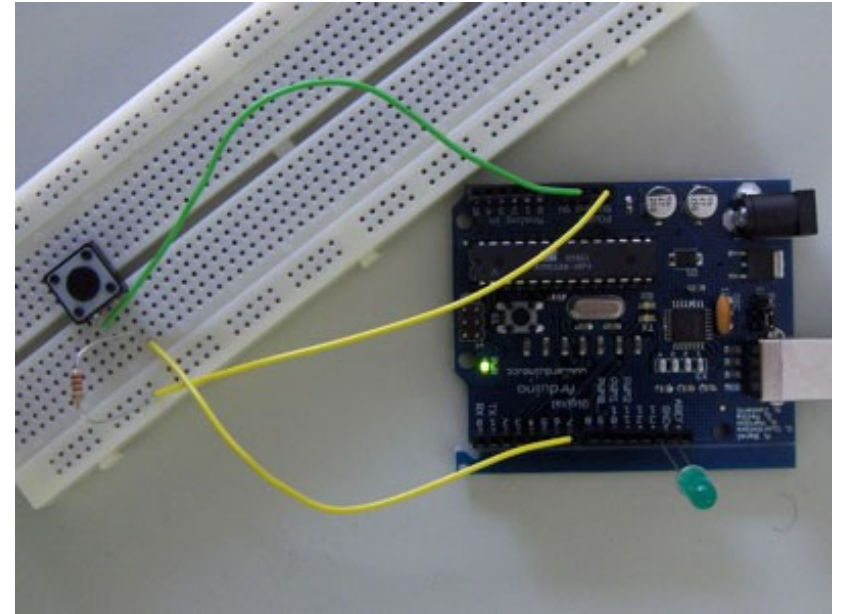
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//another example:PushButton

```
int ledPin = 13;           // choose the pin for the LED
int inputPin = 2;          // choose the input pin (for a
                           // pushbutton)
int val = 0;               // variable for reading the pin
                           // status

void setup() {
  pinMode(ledPin, OUTPUT); // declare LED as output
  pinMode(inputPin, INPUT); // declare pushbutton as input
}

void loop() {
  val = digitalRead(inputPin); // read input value
  if (val == HIGH) {           // check if the input is HIGH
    digitalWrite(ledPin, LOW); // turn LED OFF
  } else {
    digitalWrite(ledPin, HIGH); // turn LED ON
  }
}
```



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//Arduino/Examples

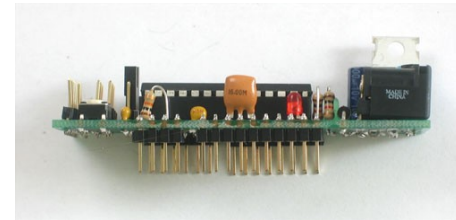
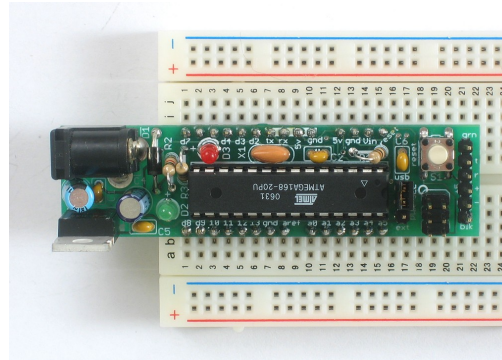
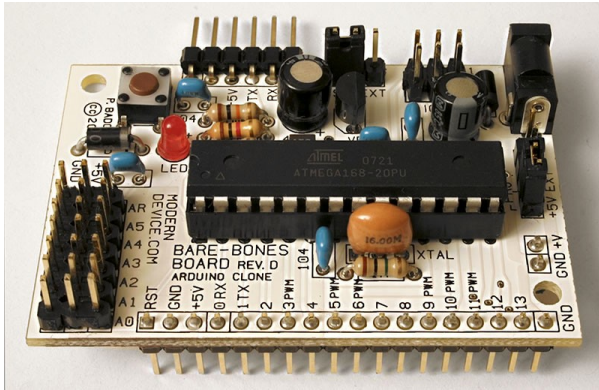
many more tutorials/examples on the web

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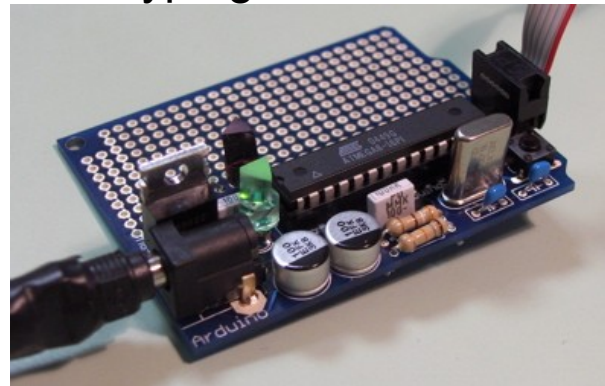
//Arduino/Related Projects/Clones

<http://moderndevise.com/>
BareBonesBoard
ReallyBareBonesBoard

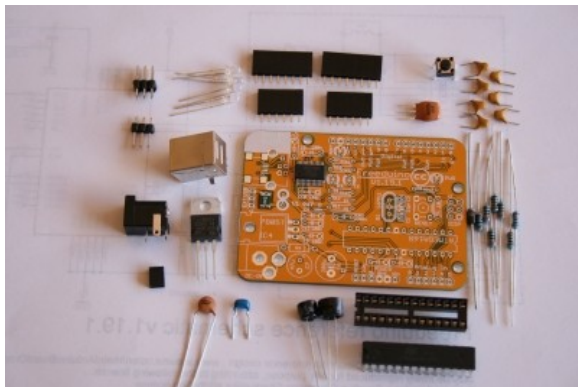
<http://www.ladyada.net/make/boarduino/>
Boarduino



<http://www.arduino.cc/playground/Main/PrototypingBoard>
PrototypingBoard



<http://www.freeduino.org/>
Freeduino

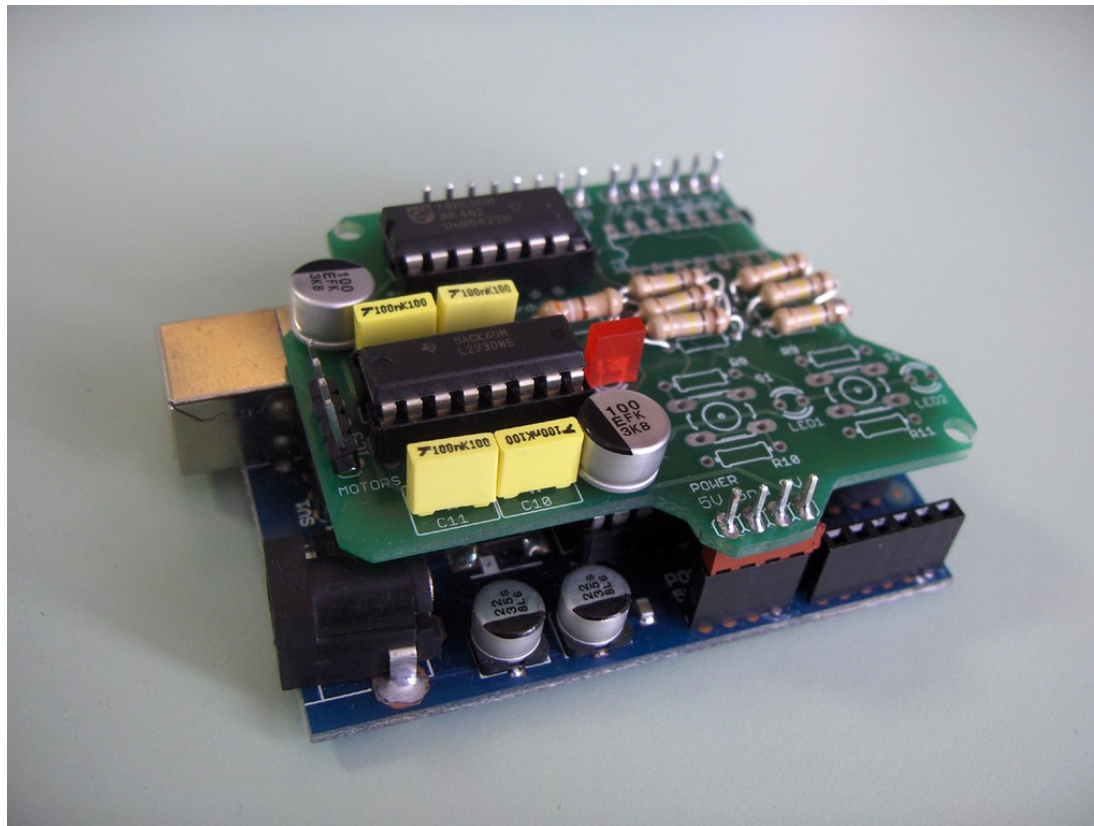


...and many more, see
<http://www.arduino.cc/playground/Main/SimilarBoards>

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//Arduino/Shields

boards that can be plugged on top of the Arduino PCB extending its capabilities



Motor Control1.1 v

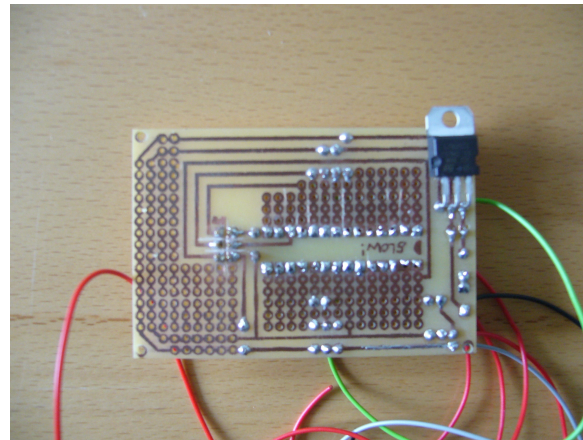
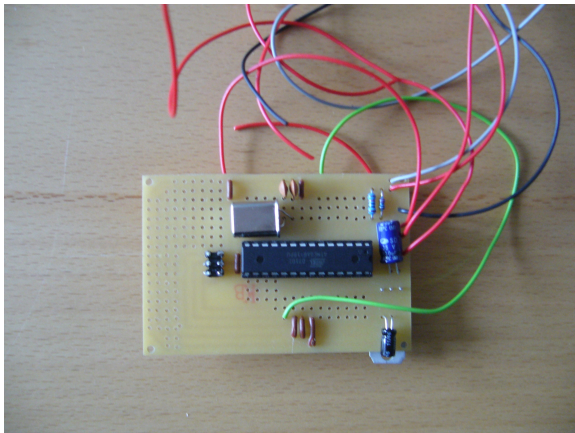
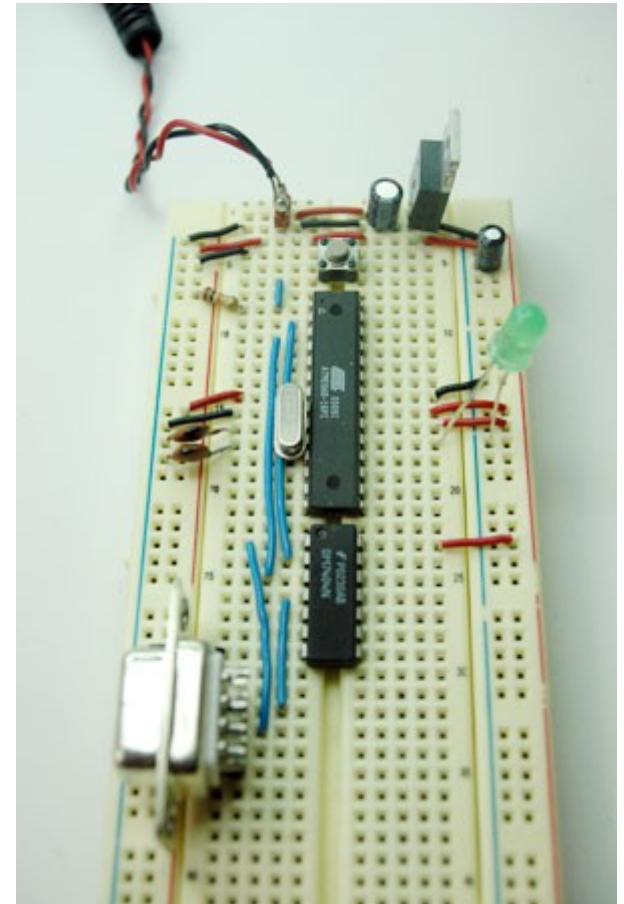
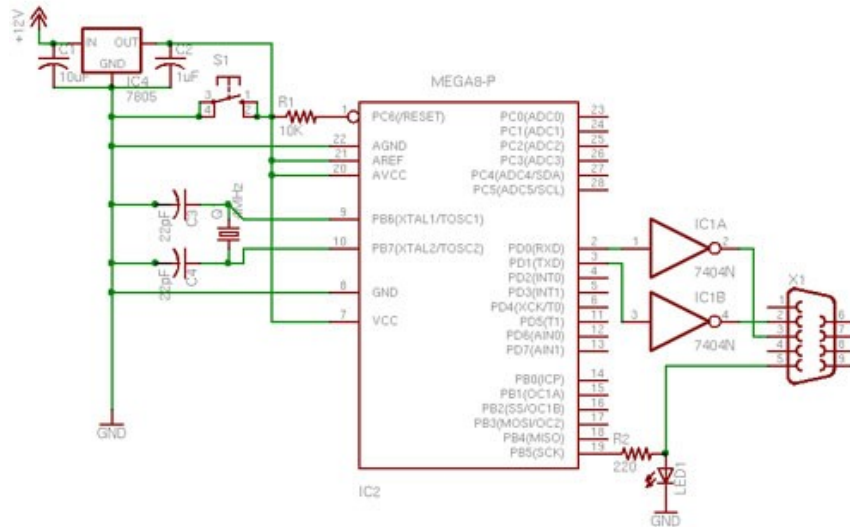
instructions how to build your own shields, eagle files, etc. can be found on the Arduino Playground:

<http://www.arduino.cc/playground/Main/SimilarBoards>

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//Arduino/Build Your Own

since the software & the hardware is open source, you can easily build your own Arduino-like board



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//tips/tricks/issues

when getting short on RAM:

- use #defines

- store constant variables/arrays in FLASH (program memory – Atmega's have seperate program and variable storage)

when getting short on FLASH

- use an ICSP (In Circuit System Programmer) and get rid of the bootloader (which will get you an other 2 KB of program memory)

send debugging info via serial port

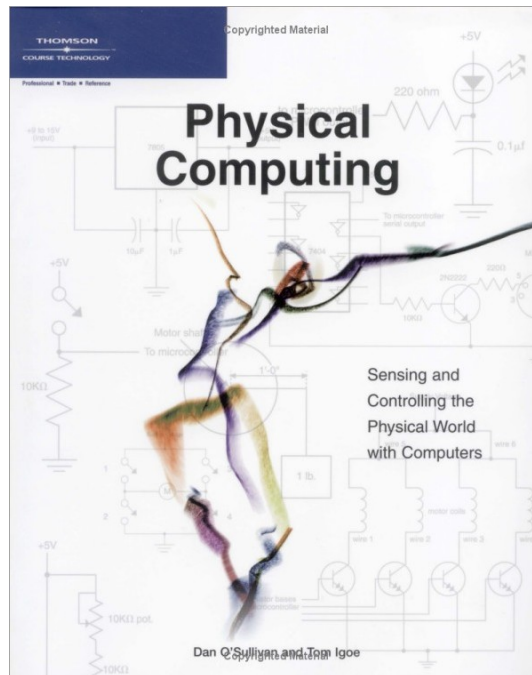
double check all your connections

check your power supply

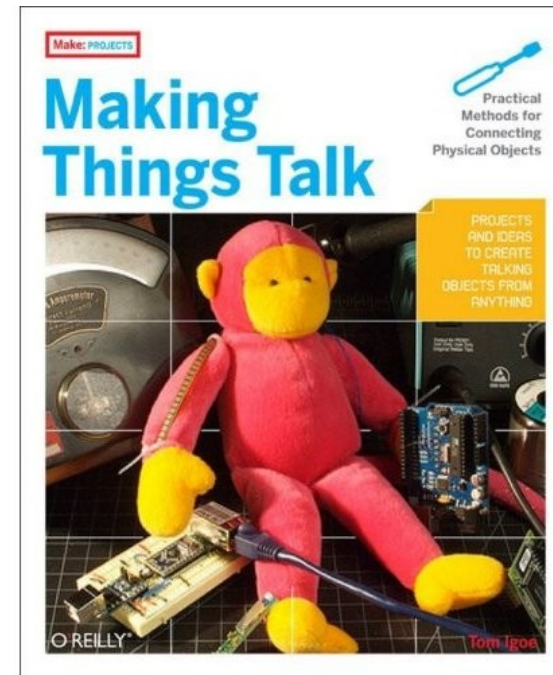
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//resources/web

- Arduino <http://www.arduino.cc/>
- Arduino Playground <http://www.arduino.cc/playground/>
- Wiring <http://wiring.org.co/>
- Processing <http://processing.org>
- AVR Libc <http://www.nongnu.org/avr-libc/>
- Tom Igoe's resources on physical computing <http://www.tigoe.net/pcomp/>
- All About Circuits <http://www.allaboutcircuits.com/>
- Socrates Electronics <http://www.ibiblio.org/kuphaldt/socratic/>



Physical Computing:
Sensing and Controlling
the Physical World with
Computers (Paperback)
by Tom Igoe, Dan
O'Sullivan



Making Things Talk:
Practical Methods for
Connecting Physical
Objects (Paperback)
by Tom Igoe (Author)

more resources on the arduino playground:
<http://www.arduino.cc/playground/Main/ElectroInfoResources>

QUESTIONS?

//3. Aufgabe

- Anwendung zur Navigation Zeit-basierter Daten
 - Potentiometer
 - in Mittelposition ($\pm$ Treshold) -> keine Translation
 - rechts/links von „Ruheposition“ -> vor / zurueck (evtl. nicht linear)
 - LED soll unterschiedlich schnell blinken, je nachdem, ob vor/zurueck navigiert wird
 - geeignetes Protokoll ueberlegen/implementieren (Arduino/Software)
- Anwendung in Processing/vvvv/C++/Java/... (frei gestellt – wenn ich es nicht innerhalb 15 Min. kompiliert bekomme, private Vorfuehrung)

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//2. Aufgabe

- Fragen zur 2. Aufgabe / vvvv
- Feedback zur 2. Aufgabe (zu leicht/zu schwierig, komische software, ... ???)
- Einige Beispiele
- Fiducial is spreadable! (example patch)

//2. Aufgabe/Alternative Zoom-Methode

zoom mittels pfeiffen

=====

mittels hohen pfeiftönen könnte man in das bild hineinzoomen, während man mit tiefen tönen herauszoomt.

vorteil:

- die hände bleiben frei

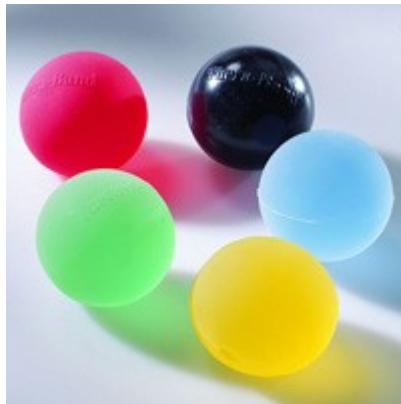
nachteil:

- die tonlagen variieren zwischen unterschiedlichen anwendern
dadurch kann es zu problemen kommen
- man kann beim bedienen kein bier trinken

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//Ausblick

- in 2 Wochen: Projekt-Proposal
- Vorschlag fuer ein Projekt:
 - analoger Druck-Sensor in elastischem Gummiball als Interface fuer verschiedene Gaming-Applications im Gesundheits/Rehabilitations-Bereich (Forschungsprojekt am Institut)
 - Prototype-Entwicklung + kleine Anwendung



- Zwischendurch noch mal ein Meeting mit allen zur Besprechung/Feedback?